


# **One code, two worlds**: building an open-source CFD code in collaboration with industry

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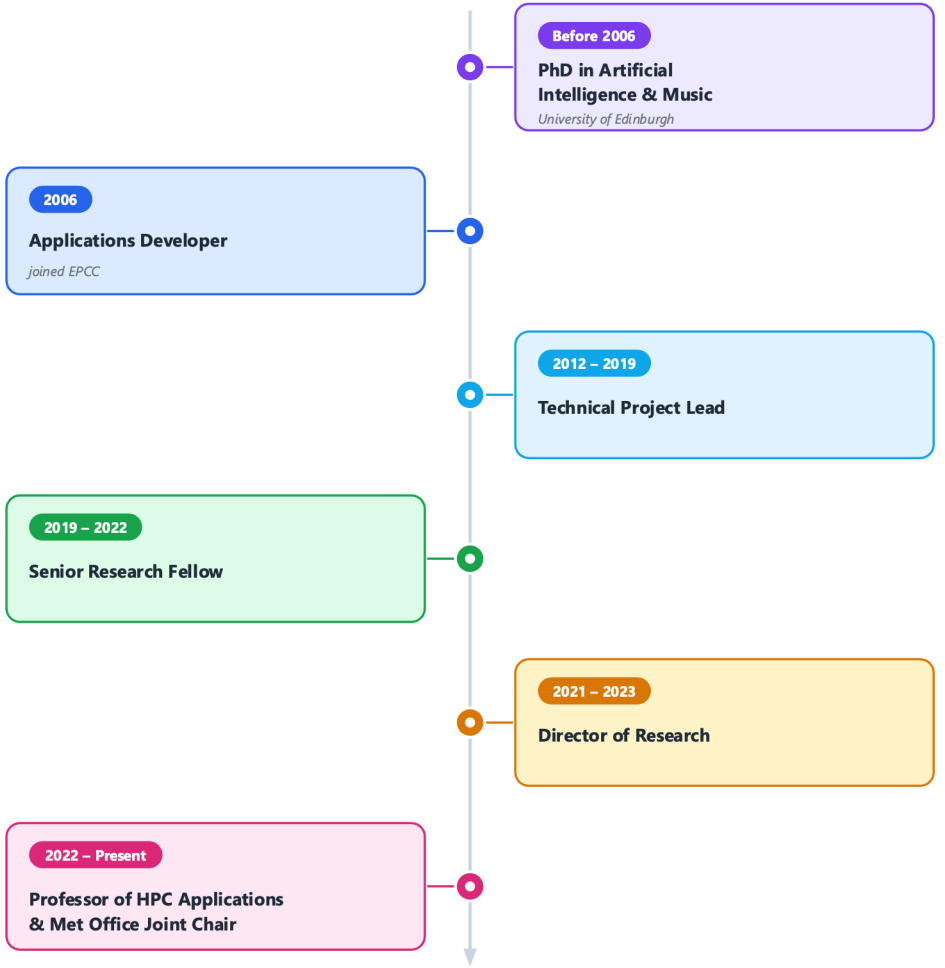
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# About me

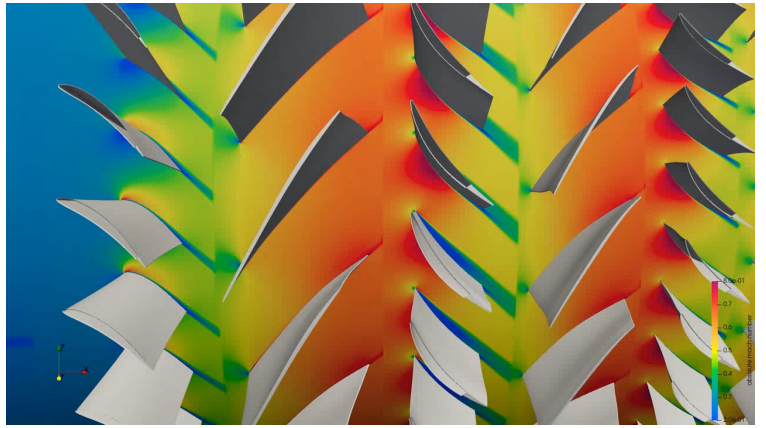
- Entire career spent at EPCC – joined straight from PhD
- Early years focused on HPC development/optimisation across a wide range of domains (from evolutionary biology to astrophysics)
- Started securing funding & publishing more
- Led to work with industry & public sector
  - Primarily Met Office, Rolls-Royce and AWE



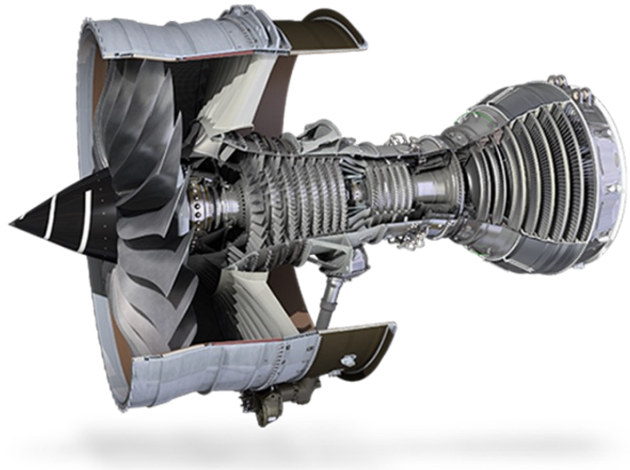
# Whole-engine simulation towards virtual certification

VECTA ASIMOV

- **Ambition**: virtual certification of aero-engines
- Requires high-fidelity multi-physics simulation of gas turbines in operation
  - Coupling CFD + structure + combustion + …
  - True Exascale problem

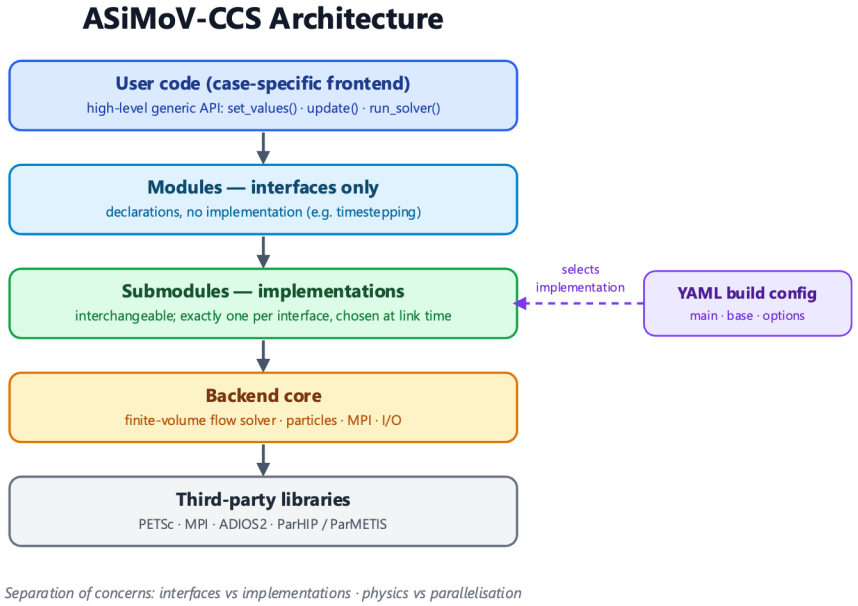


- **Key challenge**:
  - Much of the simulation code & input data is commercially sensitive
  - In-house compute resources are insufficient
- Collaboration with Rolls-Royce started 2018



# ASiMoV-CCS

- Developing alternative to Rolls-Royce's proprietary in-house CFD & combustion code to support "**grand challenge**" simulations
  - Have: Vast amounts of functionality, fully verified, validated & certified
  - Need: Improved scalability, performance, and support for GPUs
- Key requirements (in addition to the above) include
  - Improved maintainability, robustness & usability (for engineers → Fortran)
  - Backwards compatible with Rolls-Royce coupling tool & input files/meshes
  - Validated against in-house code
  - **As open as possible**



# As open as possible, as closed as necessary

- Challenge: Rolls-Royce IP is export controlled, even though large parts of code are “textbook” science
- Solution: complete separation of proprietary & open
  - Interfaces (modules) & implementations (submodules)
- Interfaces must be publishable
- Relevant implementation is chosen at link time
  - Proprietary code is kept separate from open code *at all times*!
- Key advantage is that much of the code is sharable
  - https://github.com/asimovpp/asimov-ccs/tree/develop

